Lab 4: VibeCoding - Full Chat History

Tool: Claude Code (Claude Opus 5.5), desktop app | **Student:** Pranav, PES1UG24CS466

Repo worked on: SETAPESU26/35_qbert | **Pushed to:** Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466 (Lab-4)

Session date: 2026-11-05 | **Session id:** fd18d233-152a-403c-8ec0-4d28662df9a4

Exported from the saved Claude Code session transcript. It includes every prompt, every written reply from Claude, and every action Claude took with its input and output, in order. Claude's internal reasoning is not part of the chat and is left out. Email addresses are removed, and tool outputs longer than 60 lines are shortened, with the number of lines left out shown.

Student

```
@"D:\College\sem 5\SE_mini_project\Lab_4_VibeCoding_Student_handout.pdf"
check it this is already implemented or do i need to do this and the reference lab to which u need to commit to is
https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466
and for the lab word use https://github.com/SETAPESU26/35_qbert tell me if is finished if it is finished push it to
the repo otherwise finish it and then commit it
```

Claude

I'll start by reading the handout and checking what's already in `Lab4/`, plus the state of both repos.

Action: Update task list - Read handout and inspect Lab4 folder; Check both GitHub repos and remotes; Compare requirements against existing work; Finish anything missing; Commit and push to the right repos

```
Result:
Plan shown (5 steps).
```

Action: Read - D:\College\sem 5\SE_mini_project\Lab_4_VibeCoding_Student_handout.pdf

```
Result:
pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-utils`) to
enable PDF page rendering.
```

Action: Bash - Show remotes, branches, and Lab4 file listing

```
cd "/d/College/sem 5/SE_mini_project" && git remote -v && git branch -a && git log --oneline -5 && echo '--- tree ---
' && ls -la && echo '--- Lab4 ---' && find Lab4 -path '*/node_modules' -prune -o -path '*/.git' -prune -o -type f -
print | head -200 && echo '--- size ---' && du -sh Lab4 && find Lab4 -name .git -maxdepth 4
```

```
Result:
origin    https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466.git (fetch)
origin    https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466.git (push)
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/main
aca8600 Add Lab 2 and Lab 3 deliverables: Agile backlog & sprint simulation, UML component diagram, and architectural
justification
753ccc9 Add exception flow diagram and documentation for UC-002
78fe42c Update use_case_flow_specification.pdf
60cb3d6 Update README.md with lab overview and deliverables
796ca73 Keep only required lab deliverables and problem statement PDF
--- tree ---
total 9312
drwxr-xr-x 1 sripr 197609       0 Nov  4 20:13 .
drwxr-xr-x 1 sripr 197609       0 Nov  4 11:38 ..
drwxr-xr-x 1 sripr 197609       0 Oct  7 23:34 .git
-rw-r--r-- 1 sripr 197609   68232 Aug 25 13:33 35_SE_Lab1_SE_Problem_Statements.pdf
-rw-r--r-- 1 sripr 197609  218389 Aug 30 22:29 Exception_Flow_Diagram.png
-rw-r--r-- 1 sripr 197609  158311 Aug 30 22:30 Exception_Flow_Documentation.pdf
drwxr-xr-x 1 sripr 197609       0 Oct  7 23:28 Lab2
drwxr-xr-x 1 sripr 197609       0 Oct  7 23:28 Lab3
drwxr-xr-x 1 sripr 197609       0 Nov  4 20:13 Lab4
-rw-r--r-- 1 sripr 197609  183203 Oct  7 23:21 Lab_2_Agile_Epics_and_Backlog.pdf
-rw-r--r-- 1 sripr 197609  507835 Oct  7 23:21 Lab_2_Burndown_and_Sprint_Reflection.pdf
-rw-r--r-- 1 sripr 197609  631324 Oct  7 23:21 Lab_2_Complete_Agile_Simulation_Report.pdf
-rw-r--r-- 1 sripr 197609 4347655 Sep  6 22:13 Lab_2_Jira_Student_Handout.pdf
-rw-r--r-- 1 sripr 197609   38728 Oct  7 23:17 Lab_3_Architectural_Justification.docx
-rw-r--r-- 1 sripr 197609   95342 Oct  7 23:18 Lab_3_Architectural_Justification.pdf
-rw-r--r-- 1 sripr 197609  214435 Oct  7 20:15 Lab_3_Architecture_Student_handout.pdf
-rw-r--r-- 1 sripr 197609  426226 Oct  7 23:27 Lab_3_Complete_Architecture_Report.pdf
-rw-r--r-- 1 sripr 197609  260527 Oct  7 23:10 Lab_3_Component_Diagram.pdf
-rw-r--r-- 1 sripr 197609  443975 Oct  7 23:10 Lab_3_Component_Diagram.png
-rw-r--r-- 1 sripr 197609   71355 Nov  4 20:11 Lab_4_VibeCoding_Student_handout.pdf
-rw-r--r-- 1 sripr 197609   13371 Oct  7 23:29 README.md
-rw-r--r-- 1 sripr 197609  262415 Sep  7 19:08 Screenshot (569).png
-rw-r--r-- 1 sripr 197609  290198 Sep  7 19:13 Screenshot (573).png
-rw-r--r-- 1 sripr 197609  251740 Sep  7 19:13 Screenshot (574).png
-rw-r--r-- 1 sripr 197609  269124 Sep  7 19:14 Screenshot (575).png
-rw-r--r-- 1 sripr 197609  105663 Oct  7 23:00 jira_backlog_view.png
-rw-r--r-- 1 sripr 197609   80582 Oct  7 23:03 jira_burndown_chart_view.png
-rw-r--r-- 1 sripr 197609  103171 Aug 25 14:43 requirements_table.pdf
drwxr-xr-x 1 sripr 197609       0 Oct  7 23:31 scripts
-rw-r--r-- 1 sripr 197609  194488 Oct  7 22:58 sprint1_burndown_chart.png
```

```
-rw-r--r-- 1 sripr 197609  192572 Aug 25 14:40 use_case_diagram.png
-rw-r--r-- 1 sripr 197609   25527 Aug 25 15:07 use_case_flow_specification.pdf
--- Lab4 ---
Lab4/35_qbert/.gitignore
Lab4/35_qbert/game.py
Lab4/35_qbert/README.md
--- size ---
102K    Lab4
Lab4/35_qbert/.git
```

Action: Bash - Check gh auth and inspect both GitHub repos

```
gh auth status 2>&1 | head -8; echo '--- LAB1 repo ---'; gh repo view Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466 --
json name,defaultBranchRef,viewerPermission,pushedAt 2>&1; gh api repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/
contents 2>&1 | head -c 1500; echo; echo '--- qbert repo ---'; gh repo view SETAPESU26/35_qbert --json name,
defaultBranchRef,viewerPermission,pushedAt,isPrivate 2>&1; gh api repos/SETAPESU26/35_qbert/contents --jq '.[] | .
type + " " + .path' 2>&1 | head -60; gh api 'repos/SETAPESU26/35_qbert/commits?per_page=10' --jq '.[] | .sha[0:7] + "
" + (.commit.author.name) + " " + (.commit.message | split("\n")[0])' 2>&1 | head -20
```

```
Result:
github.com
    Logged in to github.com account Pranav-146 (keyring)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_**********************************
  - Token scopes: 'gist', 'read:org', 'repo', 'workflow'
--- LAB1 repo ---
{"defaultBranchRef":{"name":"main"},"name":"LAB1_Activity_Pranav_PES1UG24CS466","pushedAt":"2026-10-07T18:04:45Z",
"viewerPermission":"ADMIN"}
[{"name":"35_SE_Lab1_SE_Problem_Statements.pdf","path":"35_SE_Lab1_SE_Problem_Statements.pdf","sha":
"d1fb0d33653629e56e3648f1c3dbf86ee1813262","size":68232,"url":"https://api.github.com/repos/Pranav-146/
LAB1_Activity_Pranav_PES1UG24CS466/contents/35_SE_Lab1_SE_Problem_Statements.pdf?ref=main","html_url":"https://github.
com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/blob/main/35_SE_Lab1_SE_Problem_Statements.pdf","git_url":"https://
api.github.com/repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/git/blobs/
d1fb0d33653629e56e3648f1c3dbf86ee1813262","download_url":"https://raw.githubusercontent.com/Pranav-146/
LAB1_Activity_Pranav_PES1UG24CS466/main/35_SE_Lab1_SE_Problem_Statements.pdf","type":"file","_links":{"self":"https://
api.github.com/repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/contents/35_SE_Lab1_SE_Problem_Statements.
pdf?ref=main","git":"https://api.github.com/repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/git/blobs/
d1fb0d33653629e56e3648f1c3dbf86ee1813262","html":"https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/
blob/main/35_SE_Lab1_SE_Problem_Statements.pdf"}},{"name":"Exception_Flow_Diagram.png","path":"Exception_Flow_Diagram.
png","sha":"b124542eae88a04da47a9abe9433be9fe6573c12","size":218389,"url":"https://api.github.com/repos/Pranav-146/
LAB1_Activity_Pranav_PES1UG24CS466/contents/Exception_Flow_Diagram.png?ref=main","html_url":"https://github.com/
Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/blob/main/Exception_Flow_Diagram.png","git_
--- qbert repo ---
{"defaultBranchRef":{"name":"main"},"isPrivate":false,"name":"35_qbert","pushedAt":"2026-09-24T09:02:34Z",
"viewerPermission":"READ"}
file .gitignore
file README.md
file game.py
bb85bb6 Vishwambhara R Hebbalalu edit readme for submission reqs
e8a8390 Vishwambhara R Hebbalalu push qbert
```

Action: Bash - Extract handout text with a Python PDF library

```
cd "/d/College/sem 5/SE_mini_project" && python -c "
import importlib
for m in ('pypdf','PyPDF2','fitz','pdfplumber'):
    try:
        importlib.import_module(m); print('have',m)
    except Exception: print('no',m)
"; python - <<'EOF'
try:
    import pypdf
    r=pypdf.PdfReader('Lab_4_VibeCoding_Student_handout.pdf')
except ImportError:
    try:
        import PyPDF2 as pypdf
        r=pypdf.PdfReader('Lab_4_VibeCoding_Student_handout.pdf')
    except ImportError:
        import fitz
        d=fitz.open('Lab_4_VibeCoding_Student_handout.pdf')
        for i,p in enumerate(d): print('=== page',i+1); print(p.get_text())
        raise SystemExit
for i,p in enumerate(r.pages):
    print('=== page',i+1); print(p.extract_text())
EOF
```

```
Result:
Exit code 1
have pypdf
have PyPDF2
no fitz
no pdfplumber
=== page 1
Traceback (most recent call last):
  File "<stdin>", line 14, in <module>
  File "C:\Users\sripr\AppData\Local\Programs\Python\Python312\Lib\encodings\cp1252.py", line 19, in encode
    return codecs.charmap_encode(input,self.errors,encoding_table)[0]
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
UnicodeEncodeError: 'charmap' codec can't encode character '\u25cf' in position 541: character maps to <undefined>
```

Action: Bash - Inspect local qbert clone history and README

```
cd "/d/College/sem 5/SE_mini_project/Lab4/35_qbert" && git remote -v && git status -sb && git log --format='%h %an <%ae> %ad %s' --date=short -15 && git branch -a && echo '--- fetch ---' && git fetch origin 2>&1 | tail -3; git status -sb | head -5; echo '--- wc ---'; wc -l game.py README.md .gitignore; echo '--- README ---'; cat README.md; echo '--- gitignore ---'; cat .gitignore
```

```
Result:
origin    https://github.com/SETAPESU26/35_qbert.git (fetch)
origin    https://github.com/SETAPESU26/35_qbert.git (push)
## main...origin/main
bb85bb6 Vishwambhara R Hebbalalu <[email removed]> 2026-09-24 edit readme for submission reqs
e8a8390 Vishwambhara R Hebbalalu <[email removed]> 2026-09-24 push qbert
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/main
--- fetch ---
## main...origin/main
--- wc ---
  228 game.py
   91 README.md
    1 .gitignore
  320 total
--- README ---
# Q*bert Repair Lab

This project is a single-file Q*bert-lite clone using **Pygame**. It introduces students to isometric projection,
diagonal hop validation, and enemy chase behavior using a small, readable object-oriented codebase.

---

## What's Provided

A working Q*bert-lite game with:

- A pyramid of cubes the player hops across diagonally, painting each cube toward its target color on landing
- Coily, an enemy that chases the player across the pyramid, and red balls that roll downward
- Falling off the edge of the pyramid (or bumping an enemy while grounded) costs a life
- Levels, lives, and scoring, with a win once every cube reaches its target color

It has **one deliberate bug** and **three optional features** left as empty functions. You are expected to
**analyze**, **interact with an AI assistant**, and **complete/fix** the game to make it fully functional and more
interesting.

### **Use an LLM (e.g. ChatGPT or Claude) as your debugging and pair-programming partner for this lab.**

---

## Getting Started

### Setup

1. Make sure you have Python 3.10+ installed.
2. Install dependencies:

```bash
pip install pygame
```

3. Run the game:

```bash
python game.py
```

Controls: Left/Up/Down/Right to hop diagonally, `R` to reset, Space to continue after clearing a level.

Tasks to Complete
[... 50 more lines of output not shown ...]
```

Action: Bash - Check branches, PRs, forks, and permissions on qbert repo

```
gh api 'repos/SETAPESU26/35_qbert/branches' --jq '.[].name' 2>&1; echo '--- PRs ---'; gh pr list --repo SETAPESU26/35_qbert --state all 2>&1 | head; echo '--- collaborators/forks ---'; gh api 'repos/SETAPESU26/35_qbert/forks' --jq '.[].full_name' 2>&1 | head; gh api 'repos/SETAPESU26/35_qbert' --jq '{permissions, allow_forking, has_issues}' 2>&1
```

```
Result:
main
--- PRs ---
1    Task 1-4 Submission - [Sakthevel Saravanan /PES1UG24CS405]    sakthevelsaravanan:main    OPEN    2026-10-08T07:03:44Z
--- collaborators/forks ---
aishwaryachila/PES2UG24CS039_SE_LAB
Vishwas-H-sys/35_qbert
Mueez751/35_qbert
```

```
sakthevelsaravanan/35_qbert
Vedaraj-HR/35_qbert
ayaan193/35_qbert
SudarshanG9/lab4-se
PavanENaik/35_qbert
niteshharsur/35_qbert
Suryatej-prog/35_qbert
{"allow_forking":true,"has_issues":true,"permissions":{"admin":false,"maintain":false,"pull":true,"push":false,
"triage":false}}
```

Action: Bash - Extract handout text using UTF-8 output

```
cd "/d/College/sem 5/SE_mini_project" && PYTHONIOENCODING=utf-8 python -c "
import pypdf
r=pypdf.PdfReader('Lab_4_VibeCoding_Student_handout.pdf')
for i,p in enumerate(r.pages):
    print('=== page',i+1); print(p.extract_text())
"

Result:
=== page 1
Lab  4:  VibeCoding
Objective
Use  Vibe  Coding  tools  to  fix  the  broken  code  and  add  features  to  the  same.  Learn  how
to

prompt

to

get

the

job

done

in

under

three

to

four

attempts.

Clone

a

repo

assigned

to

you,

make

changes

and

push

to

your

repo

in

your
[... 266 more lines of output not shown ...]
```

Action: Read - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py

```
Result:
1    import random
2    import pygame
3
4    WIDTH, HEIGHT, CUBE_W, CUBE_H = 720, 560, 72, 42
5    SIDE = 21
```

```
6    ROWS, TARGET = 7, 2
7    HOP_TIME, HOP_HEIGHT = 0.28, 26
8    CELLS = {(r, c) for r in range(ROWS) for c in range(r + 1)}
9    DEFAULT_PALETTE = [(90, 160, 220), (190, 120, 70), (100, 210, 140)]
10    KEY_HOPS = {pygame.K_LEFT: (-1, -1), pygame.K_UP: (-1, 0), pygame.K_DOWN: (1, 0), pygame.K_RIGHT: (1, 1)}
11
12
13    def cube_palette(level):
14        """Return a list of TARGET + 1 (r, g, b) colours for the cube stages, or None for the default."""
15        pass
16
17
18    def on_cube_completed(cell):
19        """Called when a cube first reaches its target colour; add a flash, sound, or bonus here."""
20        pass
21
22
23    def bonus_life_threshold():
24        """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
25        pass
26
27
28    def cube_center(row, col):
29        return pygame.Vector2(WIDTH / 2 + (col - row // 2) * CUBE_W, 90 + row * CUBE_H)
30
31
32    def neighbors(row, col):
33        return [(row - 1, col - 1), (row - 1, col), (row + 1, col), (row + 1, col + 1)]
34
35
36    def shade(color, factor):
37        return tuple(int(v * factor) for v in color)
38
39
40    class Hopper:
41        def __init__(self, cell):
42            self.cell, self.target, self.t, self.fall = cell, None, 0.0, None
43
44        @property
45        def busy(self):
46            return self.target is not None or self.fall is not None
47
48        def start(self, target):
49            self.target, self.t = target, 0.0
50
51        def update(self, dt):
52            """Advance the hop; return the landing cell when a hop finishes on the pyramid."""
53            if self.fall is not None:
54                self.fall += 420 * dt
55                return None
56            if self.target is None:
57                return None
58            self.t += dt / HOP_TIME
59            if self.t < 1:
[... 170 more lines of output not shown ...]
```

Action: Bash - See how the existing PR submits and check pygame

```
gh pr view 1 --repo SETAPESU26/35_qbert --json title,body,files,author --jq '{title, author: .author.login, body,
files: [.files[].path]}' 2>&1 | head -40; python -c "import pygame; print(pygame.__version__)" 2>&1 | tail -1
```

```
Result:
{"author":"sakthevelsaravanan","body":"","files":["game.py"],"title":"Task 1-4 Submission - [Sakthevel Saravanan /
PES1UG24CS405]"}
ModuleNotFoundError: No module named 'pygame'
```

Action: Update task list - Read handout and inspect Lab4 folder; Check both GitHub repos and remotes; Compare requirements against existing work; Record before video from original code; Fix projection bug and add three features; Test game headless and record after video; Assemble Lab-4 folder and export chat history; Commit and push to personal repo

```
Result:
Plan shown (8 steps).
```

Action: Bash - Review repo layout conventions and README structure

```
cd "/d/College/sem 5/SE_mini_project" && echo '--- tracked in Lab2/Lab3/scripts ---'; git ls-files Lab2 Lab3 scripts;
echo '--- handouts tracked ---'; git ls-files | grep -i handout; echo '--- README headings ---'; grep -n '^#' README.
md | head -50; echo '--- tail ---'; tail -15 README.md; echo '--- visibility ---'; gh repo view Pranav-146/
LAB1_Activity_Pranav_PES1UG24CS466 --json isPrivate; git fetch origin 2>&1 | tail -1; git status -sb | head -3
```

```
Result:
--- tracked in Lab2/Lab3/scripts ---
Lab2/Lab_2_Agile_Epics_and_Backlog.pdf
Lab2/Lab_2_Burndown_and_Sprint_Reflection.pdf
Lab2/Lab_2_Complete_Agile_Simulation_Report.pdf
Lab2/README.md
Lab2/Screenshot (569).png
Lab2/Screenshot (573).png
Lab2/Screenshot (574).png
```

```
Lab2/Screenshot (575).png
Lab2/jira_backlog_view.png
Lab2/jira_burndown_chart_view.png
Lab2/sprint1_burndown_chart.png
Lab3/Lab_3_Architectural_Justification.docx
Lab3/Lab_3_Architectural_Justification.pdf
Lab3/Lab_3_Complete_Architecture_Report.pdf
Lab3/Lab_3_Component_Diagram.pdf
Lab3/Lab_3_Component_Diagram.png
Lab3/README.md
scripts/generate_lab2_pdfs.py
scripts/generate_lab3_complete_report.py
scripts/generate_lab3_justification.py
--- handouts tracked ---
Lab_2_Jira_Student_Handout.pdf
Lab_3_Architecture_Student_handout.pdf
--- README headings ---
1:# Software Engineering Laboratory Portfolio: Problem Statement #35
13:##   Repository Quick Navigation
22:##   Problem Overview: Customizable Subscription Box Scheduler
29:##   Lab 1: Requirements Engineering & Use-Case Modelling
31:### Summary of Requirements
40:### Lab 1 Deliverables
55:##   Lab 2: Agile Backlog Creation & Sprint Simulation in Jira
59:### Objectives & Agile Transformation
62:### Lab 2 Deliverables
76:### Agile Epics & Stories Breakdown
88:### Burndown Chart Preview
91:### Summary Answers to Reflection Questions
99:##    Lab 3: Component Modelling & Architectural Pattern Selection
103:### Selected Architectural Style
106:### Lab 3 Deliverables
116:### UML Component Diagram Preview
119:### Summary of Architectural Justification
129:##    Repository Deliverables Index
--- tail ---
│
└── Lab 3 Files (Component Modelling & Architecture)
    ├── Lab_3_Component_Diagram.png              # Lab 3 Deliverable 1: Component Diagram PNG
    ├── Lab_3_Component_Diagram.pdf              # Lab 3 Deliverable 1: Component Diagram PDF
    ├── Lab_3_Architectural_Justification.pdf    # Lab 3 Deliverable 2: 1-Page Written Justification PDF
    ├── Lab_3_Architectural_Justification.docx   # Lab 3 Deliverable 2: 1-Page Written Justification DOCX
    ├── Lab_3_Complete_Architecture_Report.pdf   # Lab 3 Consolidated 4-Page Architecture Report
    └── Lab3/                                    # Dedicated Lab 3 Directory
        ├── README.md                            # Comprehensive Lab 3 documentation
        ├── Lab_3_Component_Diagram.png
        ├── Lab_3_Component_Diagram.pdf
        ├── Lab_3_Architectural_Justification.pdf
        ├── Lab_3_Architectural_Justification.docx
        └── Lab_3_Complete_Architecture_Report.pdf
```
[... 5 more lines of output not shown ...]

Action: Bash - Create scratch venv with game and video packages

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; python
-m venv "$S/venv" && "$S/venv/Scripts/python.exe" -m pip install -q pygame imageio imageio-ffmpeg reportlab 2>&1 |
tail -3; "$S/venv/Scripts/python.exe" -c "import pygame,imageio,imageio_ffmpeg,reportlab;print(pygame.__version__,
imageio.__version__, reportlab.__version__)" | tail -1; cp "/d/College/sem 5/SE_mini_project/Lab4/35_qbert/game.py"
"$S/game_before.py"; ls "$S"
```

```
Result:
[notice] A new release of pip is available: 23.2.1 -> 26.3
[notice] To update, run: C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-
4d28662df9a4\scratchpad\venv\Scripts\python.exe -m pip install --upgrade pip
2.6.1 2.38.1 5.1.0
game_before.py
venv
```

Action: ToolSearch

```
{
 "query": "select:mcp__ccd_session_mgmt__export_transcript",
 "max_results": 1
}
```

```
Result:
(no output)
```

Action: Edit - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py

```
- KEY_HOPS = {pygame.K_LEFT: (-1, -1), pygame.K_UP: (-1, 0), pygame.K_DOWN: (1, 0), pygame.K_RIGHT: (1, 1)}
-
-
- def cube_palette(level):
-     """Return a list of TARGET + 1 (r, g, b) colours for the cube stages, or None for the default."""
-     pass
-
-
- def on_cube_completed(cell):
```

```
-     """Called when a cube first reaches its target colour; add a flash, sound, or bonus here."""
-     pass
-
-
- def bonus_life_threshold():
-     """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
-     pass
-
-
- def cube_center(row, col):
-     return pygame.Vector2(WIDTH / 2 + (col - row // 2) * CUBE_W, 90 + row * CUBE_H)
+ KEY_HOPS = {pygame.K_LEFT: (-1, -1), pygame.K_UP: (-1, 0), pygame.K_DOWN: (1, 0), pygame.K_RIGHT: (1, 1)}
+ PALETTES = [
+     DEFAULT_PALETTE,
+     [(150, 90, 200), (235, 120, 170), (255, 225, 110)],
+     [(200, 80, 70), (240, 160, 60), (130, 230, 220)],
+     [(70, 130, 110), (170, 200, 80), (245, 245, 200)],
+ ]
+ FLASH_TIME = 0.35
+ flashes = {}
+
+
+ def cube_palette(level):
+     """Return a list of TARGET + 1 (r, g, b) colours for the cube stages, or None for the default."""
+     return PALETTES[(level - 1) % len(PALETTES)]
+
+
+ def on_cube_completed(cell):
+     """Called when a cube first reaches its target colour; add a flash, sound, or bonus here."""
+     flashes[cell] = FLASH_TIME
+
+
+ def bonus_life_threshold():
+     """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
+     return 1000
+
+
+ def cube_center(row, col):
+     return pygame.Vector2(WIDTH / 2 + (col - row / 2) * CUBE_W, 90 + row * CUBE_H)
```

Result:
The file D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py has been updated successfully. (file state is current in your context — no need to Read it back)

Action: Edit - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py

```
-         self.stages = {cell: 0 for cell in CELLS}
-         self.state = "play"
+         self.stages = {cell: 0 for cell in CELLS}
+         flashes.clear()
+         self.state = "play"
```

Result:
The file D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py has been updated successfully. (file state is current in your context — no need to Read it back)

Action: Edit - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py

```
-     def update(self, dt):
-         if self.state != "play":
-             return
-         landed = self.player.update(dt)
+     def update(self, dt):
+         for cell in list(flashes):
+             flashes[cell] -= dt
+             if flashes[cell] <= 0:
+                 del flashes[cell]
+         if self.state != "play":
+             return
+         landed = self.player.update(dt)
```

Result:
The file D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py has been updated successfully. (file state is current in your context — no need to Read it back)

Action: Edit - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py

```
-         color = colors[self.stages[cell]]
-         left =
+         color = colors[self.stages[cell]]
+         if cell in flashes:
+             glow = flashes[cell] / FLASH_TIME
+             color = tuple(int(v + (255 - v) * glow) for v in color)
+         left =
```

Result:
The file D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py has been updated successfully. (file state is current in your context — no need to Read it back)

Action: Write - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\record.py

```
"""Drive the real game code headlessly with a simple bot and record frames.

usage: record.py <game.py> <out.mp4> <before|after>
Frames come from Game.draw(); inputs go through Game.hop() exactly as main() does.
"""
import collections
import importlib.util
import os
import random
import sys

os.environ["SDL_VIDEODRIVER"] = "dummy"
os.environ["SDL_AUDIODRIVER"] = "dummy"
import imageio.v2 as imageio
import pygame

game_path, out_path, mode = sys.argv[1:4]
spec = importlib.util.spec_from_file_location("game_under_test", game_path)
g = importlib.util.module_from_spec(spec)
spec.loader.exec_module(g)

FPS, SECONDS, DT = 60, 10, 1 / 60
pygame.init()
screen = pygame.display.set_mode((g.WIDTH, g.HEIGHT))
HOPS = {v: k for k, v in g.KEY_HOPS.items()}


def danger(game):
    cells = set()
    enemies = ([game.coily] if game.coily else []) + game.balls
    for e in enemies:
        if e.fall is not None:
            continue
        spots = [e.cell] + ([e.target] if e.target else [])
        for s in spots:
            cells.add(s)
            cells.update(g.neighbors(*s))
    return cells


def choose(game, rng):
    here = game.player.cell
    bad = danger(game)
    todo = {c for c, s in game.stages.items() if s < g.TARGET}
    queue = collections.deque([(here, None)])
    seen = {here}
    while queue:
        cell, first = queue.popleft()
        if first is not None and cell in todo:
            return first
        for n in g.neighbors(*cell):
            if n in g.CELLS and n not in seen and n not in bad:
                seen.add(n)
                queue.append((n, first or n))
    safe = [n for n in g.neighbors(*here) if n in g.CELLS and n not in bad]
    options = safe or [n for n in g.neighbors(*here) if n in g.CELLS]
    return rng.choice(options)


def run(seed, on_frame=None, limit=90 * FPS):
    random.seed(seed)
    rng = random.Random(seed)
    game = g.Game()
    log = []
    wait = 0
    for frame in range(limit):
        if game.state == "play" and not game.player.busy:
            nxt = choose(game, rng)
            here = game.player.cell
            game.hop(HOPS[(nxt[0] - here[0], nxt[1] - here[1])])
        if game.state == "win":
            wait += 1
            if wait > 45:
                wait = 0
                game.level += 1
                game.reset(full=False)
        game.update(DT)
        log.append((game.score, game.lives, game.level, game.state))
        if on_frame:
            on_frame(frame, game)
        if game.state == "lose":
            break
    return log
```

```
def window_for(log):
    """10s window containing the 1000-point bonus, the level clear and level 2."""
    cross = next((i for i, r in enumerate(log) if r[0] >= 1000), None)
    lvl2 = next((i for i, r in enumerate(log) if r[2] == 2), None)
    if cross is None or lvl2 is None:
        return None
    end = lvl2 + 90
    start = end - SECONDS * FPS
    return (start, end) if start >= 0 and start <= cross - 60 and end < len(log) else None


if mode == "after":
    for seed in range(5000):
        win = window_for(run(seed))
        if win:
            break
    else:
        sys.exit("no seed produced a suitable window")
else:
    seed, win = 1, (0, SECONDS * FPS)
print("seed", seed, "window", win)

writer = imageio.get_writer(out_path, fps=30, codec="libx264", quality=8, macro_block_size=8)


def grab(frame, game):
    if win[0] <= frame < win[1] and frame % 2 == 0:
        game.draw(screen)
        writer.append_data(pygame.surfarray.array3d(screen).swapaxes(0, 1))
        if frame in (win[0], (win[0] + win[1]) // 2, win[1] - 2):
            pygame.image.save(screen, f"{out_path}.{frame}.png")


log = run(seed, grab)
writer.close()
print("frames logged", len(log), "start", log[win[0]], "end", log[min(win[1], len(log)) - 1])
```

Result:
File created successfully at: C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\record.py (file state is current in your context — no need to Read it back)

Action: Write - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\check_game.py

```
"""Headless checks of the fixed game against the README's expected behaviour."""
import importlib.util
import os
import sys

os.environ["SDL_VIDEODRIVER"] = "dummy"
import pygame

spec = importlib.util.spec_from_file_location("g", sys.argv[1])
g = importlib.util.module_from_spec(spec)
spec.loader.exec_module(g)
pygame.init()
screen = pygame.display.set_mode((g.WIDTH, g.HEIGHT))
DT = 1 / 60


def fresh():
    game = g.Game()
    game.coily_timer = game.ball_timer = 1e9
    return game


def settle(game):
    for _ in range(40):
        game.update(DT)
        if not game.player.busy:
            break


# Task 1: every row symmetric about the centre line, constant step
for r in range(g.ROWS):
    xs = [g.cube_center(r, c).x for c in range(r + 1)]
    assert abs((xs[0] + xs[-1]) / 2 - g.WIDTH / 2) < 1e-9, (r, xs)
print("ok  pyramid symmetric on all rows")

# Hop directions are true diagonals
for key, name, sx, sy in [(pygame.K_RIGHT, "down-right", 1, 1), (pygame.K_DOWN, "down-left", -1, 1)]:
    d = g.KEY_HOPS[key]
    v = g.cube_center(2 + d[0], 1 + d[1]) - g.cube_center(2, 1)
    assert v.x * sx > 0 and v.y * sy > 0, name
print("ok  hops are diagonal")

# Task 2: length, per-level difference, wraps, valid rgb
```

```
pals = [g.cube_palette(n) for n in range(1, 9)]
assert all(len(p) == g.TARGET + 1 for p in pals)
assert all(0 <= v <= 255 for p in pals for c in p for v in c)
assert len({str(p) for p in pals[:4]}) == 4 and pals[0] == pals[4]
print("ok  palettes valid, differ per level, wrap")

# One stage per hop; Task 3 fires only on the completing hop
calls = []
real = g.on_cube_completed
g.on_cube_completed = lambda cell: (calls.append(cell), real(cell))
game = fresh()
game.hop(pygame.K_DOWN); settle(game)
assert game.stages[(1, 0)] == 1 and not calls and game.score == 25
game.hop(pygame.K_UP); settle(game)
game.hop(pygame.K_DOWN); settle(game)
assert game.stages[(1, 0)] == 2 and calls == [(1, 0)]
assert (1, 0) in g.flashes
game.draw(screen)
game.hop(pygame.K_UP); settle(game)
game.hop(pygame.K_DOWN); settle(game)
assert game.stages[(1, 0)] == 2 and calls == [(1, 0)]
print("ok  one stage per hop, completion hook fires once")

# Flash expires, and also expires when the game is not in "play"
for _ in range(40):
    game.update(DT)
assert not g.flashes
g.on_cube_completed((3, 1))
game.state = "win"
for _ in range(40):
    game.update(DT)
assert not g.flashes
print("ok  flash expires, including on the level-clear screen")

# Flash really changes the pixel
game = fresh()
cx, cy = map(int, g.cube_center(3, 1))
game.draw(screen); plain = screen.get_at((cx + 20, cy))[:3]
g.on_cube_completed((3, 1))
game.draw(screen); lit = screen.get_at((cx + 20, cy))[:3]
assert lit != plain and sum(lit) > sum(plain)
print("ok  flash visible on screen", plain, "->", lit)

# Task 4: one life per multiple, not repeated
game = fresh()
game.score = 990; game.update(DT); assert game.lives == 3
game.score = 1000; game.update(DT); assert game.lives == 4
game.update(DT); assert game.lives == 4
game.score = 2100; game.update(DT); assert game.lives == 5
print("ok  bonus life once per 1000")

# Falling costs a life and respawns
game = fresh()
game.hop(pygame.K_LEFT)
for _ in range(120):
    game.update(DT)
assert game.lives == 2 and game.player.cell == (0, 0) and game.state == "play"
print("ok  falling costs a life")

# Coily moves toward the player
game = fresh()
game.player.cell = (6, 6)
game.coily = g.Hopper((0, 0))
game.enemy_hop = 0
for _ in range(60):
    game.player.target = None
    game.update(DT)
assert game.coily.cell[0] > 0 and game.coily.cell[1] > 0
print("ok  Coily chases", game.coily.cell)

# Win and lose states
game = fresh()
game.stages = {c: g.TARGET for c in g.CELLS}
game.update(DT)
assert game.state == "win"
game.level += 1; game.reset(full=False)
assert game.level == 2 and game.state == "play" and not any(game.stages.values())
game = fresh(); game.lives = 1; game.lose_life()
assert game.state == "lose"
print("ok  win, next level, and game over")
print("ALL CHECKS PASSED")
```

Result:
File created successfully at: C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\check_game.py (file state is current in your context — no need to Read it back)

Action: Bash - Run game checks and record both videos

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; G="/d/
College/sem 5/SE_mini_project/Lab4/35_qbert/game.py"; PY="$S/venv/Scripts/python.exe"; mkdir -p "$S/out"; cd "$S";
echo '=== checks ==='; "$PY" -I check_game.py "$G" 2>&1 | grep -v -i 'pygame \|hello from'; echo '=== before ===';
"$PY" record.py game_before.py out/before.mp4 before 2>&1 | grep -v -i 'pygame \|hello from' | tail -5; echo '===
after ==='; "$PY" record.py "$G" out/after.mp4 after 2>&1 | grep -v -i 'pygame \|hello from' | tail -5; ls -la out
```

```
Result:
=== checks ===
ok  pyramid symmetric on all rows
ok  hops are diagonal
ok  palettes valid, differ per level, wrap
Traceback (most recent call last):
  File "C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\
check_game.py", line 64, in <module>
    assert game.stages[(1, 0)] == 2 and calls == [(1, 0)]
AssertionError
=== before ===
seed 1 window (0, 600)
frames logged 2052 start (0, 3, 1, 'play') end (800, 3, 1, 'play')
=== after ===
seed 15 window (878, 1478)
frames logged 5400 start (950, 2, 1, 'play') end (2025, 3, 2, 'play')
total 764
drwxr-xr-x 1 sripr 197609      0 Nov  5 20:08 .
drwxr-xr-x 1 sripr 197609      0 Nov  5 20:08 ..
-rw-r--r-- 1 sripr 197609 351433 Nov  5 20:08 after.mp4
-rw-r--r-- 1 sripr 197609  14443 Nov  5 20:08 after.mp4.1178.png
-rw-r--r-- 1 sripr 197609  13866 Nov  5 20:08 after.mp4.1476.png
-rw-r--r-- 1 sripr 197609  15452 Nov  5 20:08 after.mp4.878.png
-rw-r--r-- 1 sripr 197609 327044 Nov  5 20:08 before.mp4
-rw-r--r-- 1 sripr 197609  12502 Nov  5 20:08 before.mp4.0.png
-rw-r--r-- 1 sripr 197609  14867 Nov  5 20:08 before.mp4.300.png
-rw-r--r-- 1 sripr 197609  15270 Nov  5 20:08 before.mp4.598.png
Shell cwd was reset to D:\College\sem 5\SE_mini_project
```

Action: Bash - Fix check, rerun, verify durations and video timeline

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; G="/d/
College/sem 5/SE_mini_project/Lab4/35_qbert/game.py"; PY="$S/venv/Scripts/python.exe"; cd "$S"; sed -n '64p'
check_game.py; sed -i '64s/.*/assert game.stages[(1, 0)] == 2 and calls.count((1, 0)) == 1 and calls == [(1, 0), (0,
0)]/' check_game.py; "$PY" -I check_game.py "$G" 2>&1 | grep -v -i 'pygame \|hello from'; echo '=== durations ===';
"$PY" - <<'EOF'
import imageio.v2 as imageio
for f in ('out/before.mp4','out/after.mp4'):
    r=imageio.get_reader(f); m=r.get_meta_data(); print(f,m.get('duration'),m.get('fps'),m.get('size'))
EOF
echo '=== after timeline ==='; cat > tl.py <<'EOF'
import re,sys
src=open('record.py').read().split('if mode == "after":')[0]
sys.argv=['x',sys.argv[1],'x','after']
exec(src)
log=run(15)
prev=None
for i,r in enumerate(log[878:1478],878):
    k=(r[1],r[2],r[3])
    if k!=prev: print(round((i-878)/60,2),'s',r); prev=k
EOF
"$PY" tl.py "$G" 2>&1 | grep -v -i 'pygame \|hello from'
```

```
Result:
assert game.stages[(1, 0)] == 2 and calls == [(1, 0)]
ok  pyramid symmetric on all rows
ok  hops are diagonal
ok  palettes valid, differ per level, wrap
ok  one stage per hop, completion hook fires once
ok  flash expires, including on the level-clear screen
ok  flash visible on screen (90, 160, 220) -> (255, 255, 255)
ok  bonus life once per 1000
ok  falling costs a life
ok  Coily chases (2, 2)
ok  win, next level, and game over
ALL CHECKS PASSED
=== durations ===
out/before.mp4 10.0 30.0 (720, 560)
out/after.mp4 10.0 30.0 (720, 560)
We had to kill ffmpeg to stop it.
=== after timeline ===
0.0 s (950, 2, 1, 'play')
1.22 s (1000, 3, 1, 'play')
3.77 s (1200, 2, 1, 'play')
7.73 s (1900, 2, 1, 'win')
8.5 s (1900, 2, 2, 'play')
9.63 s (2000, 3, 2, 'play')
Shell cwd was reset to D:\College\sem 5\SE_mini_project
[This command modified 1 file you've previously read: C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\
fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\check_game.py. Call Read before editing.]
```

Action: Read - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\out\before.mp4.300.png

```
Result:
[image shown to Claude]
```

Action: Read - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\out\after.mp4.1178.png

```
Result:
[image shown to Claude]
```

Action: Read - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\out\after.mp4.1476.png

```
Result:
[image shown to Claude]
```

Action: Update task list - {"advance": true, "done": [4, 5, 6], "start": 7}

```
Result:
Plan updated: step 8 of 8 is in progress. App actions available now: commit, create_pr, verify.
```

Action: mcp__ccd_session_mgmt__export_transcript

```
{
 "session_id": "self"
}
```

```
Result:
Exported this session's transcript "Lab 4 VibeCoding implementation status" to C:\Users\sripr\Downloads\session-export-1793889569964.zip (754 KB, 772273 bytes). The zip holds the conversation transcript, its subagent transcripts and session metadata, not the app's logs; nothing was uploaded.
```

Action: Read - D:\College\sem 5\SE_mini_project\README.md

```
Result:
1    # Software Engineering Laboratory Portfolio: Problem Statement #35
2
3    **Course:** Software Engineering (5th Semester) &bull; Department of Computer Science &amp; Engineering &bull; PES University
4    **Student Name:** Pranav
5    **SRN:** PES1UG24CS466
6    **Problem Statement #35:** Customizable Subscription Box Scheduler
7    **Domain:** Retail, E-Commerce & Finance
8    **Jira Workspace:** [sripranavbuduguru.atlassian.net](https://sripranavbuduguru.atlassian.net/) &bull; **Project Key:** `SCRUM`
9    **GitHub Repository:** [Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466](https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466)
10
11    ---
12
13    ##   Repository Quick Navigation
14
15    - [Lab 1: Requirements Engineering &amp; Use-Case Modelling](#-lab-1-requirements-engineering--use-case-modelling)
16    - [Lab 2: Agile Backlog Creation &amp; Sprint Simulation in Jira](#-lab-2-agile-backlog-creation--sprint-simulation-in-jira)
17    - [Lab 3: Component Modelling &amp; Architectural Pattern Selection](#-lab-3-component-modelling--architectural-pattern-selection)
18    - [Full Deliverables Directory Listing](#-repository-deliverables-index)
19
20    ---
21
22    ##   Problem Overview: Customizable Subscription Box Scheduler
```

Action: Read - D:\College\sem 5\SE_mini_project\README.md

```
Result:
118
119    ### Summary of Architectural Justification
120    - **Architectural Choice:** Event-driven Microservices Architecture decomposing the system into 7 services: API Gateway, Order & Subscription Manager, Recommendation & Preference Engine, Payment Service, Fulfillment Manifest Engine, Automated Notification Service, and Clustered Storage.
121    - **Two Specific Reasons:**
122      1. *Elastic Autoscaling for 48-Hour Customization Bursts (FR-001, NFR-002):* Independent horizontal scaling of client-facing recommendation and customization pods to sustain **50,000 concurrent sessions** without over-provisioning back-office services.
123      2. *Fault Isolation Between Personalization and Warehouse Fulfillment (FR-002, FR-004):* Algorithmic delays or model retraining in the Recommendation Engine do not block scheduled recurring billing or warehouse shipping label generation.
124    - **Security Advantage:** Dedicated **PCI-DSS Level 1** isolated Payment Service using tokenization and TLS 1.3 encryption (cardholder data never touches core databases); zero-trust OAuth 2.0 / JWT API Gateway enforces strict Role-Based Access Control (RBAC).
125    - **Performance Benefit:** Asynchronous batch processing satisfies **NFR-001** (exporting **10,000 box shipping labels in ~38 seconds**, well below the 1-minute requirement) via multi-threaded workers and direct S3 cloud streaming.
126
127    ---
128
129    ##    Repository Deliverables Index
130
```

```
131    ```
132    .
133    ├── README.md                                  # Master Laboratory Portfolio (Labs 1, 2, and 3)
134    │
135    ├── Lab 1 Files
136    │   ├── 35_SE_Lab1_SE_Problem_Statements.pdf     # Original Problem Statement #35
137    │   ├── requirements_table.pdf                   # Lab 1 Requirements Specification (5 FRs, 2 NFRs)
```

Action: Bash - Unpack exported transcript and inspect its structure

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; mkdir -
p "$S/export" && cd "$S/export" && unzip -q -o /c/Users/sripr/Downloads/session-export-1793889569964.zip && find . -
type f | head -20 && F=$(find . -name '*.jsonl' | head -1); echo "main: $F"; wc -l "$F"; "$S/venv/Scripts/python.exe"
-I - "$F" <<'EOF'
import json,sys,collections
c=collections.Counter()
for l in open(sys.argv[1],encoding='utf-8'):
    d=json.loads(l)
    m=d.get('message') or {}
    ct=m.get('content')
    kinds=[b.get('type') for b in ct] if isinstance(ct,list) else [type(ct).__name__]
    c[(d.get('type'),m.get('role'),tuple(kinds)[:2])]+=1
for k,v in c.most_common(20): print(v,k)
EOF
```

```
Result:
./fd18d233-152a-403c-8ec0-4d28662df9a4/custom-title.json
./fd18d233-152a-403c-8ec0-4d28662df9a4.jsonl
./local-session-state.json
./metadata.json
./transcript.jsonl
main: ./fd18d233-152a-403c-8ec0-4d28662df9a4.jsonl
156 ./fd18d233-152a-403c-8ec0-4d28662df9a4.jsonl
25 ('assistant', 'assistant', ('tool_use',))
25 ('user', 'user', ('tool_result',))
20 ('attachment', None, ('NoneType',))
12 ('last-prompt', None, ('NoneType',))
12 ('atis-latch', None, ('NoneType',))
11 ('custom-title', None, ('NoneType',))
10 ('agent-name', None, ('NoneType',))
9 ('assistant', 'assistant', ('thinking',))
7 ('api-request', None, ('NoneType',))
5 ('tools-used', None, ('NoneType',))
4 ('api-request-blob', 'system', ('text',))
3 ('file-history-delta', None, ('NoneType',))
2 ('queue-operation', None, ('NoneType',))
2 ('api-request-shape', None, ('NoneType',))
2 ('api-request-blob', 'user', ('tool_result', 'tool_result'))
2 ('api-request-blob', 'assistant', ('thinking', 'thinking'))
1 ('user', 'user', ('str',))
1 ('file-history-snapshot', None, ('NoneType',))
1 ('api-request-blob', 'user', ('text', 'text'))
1 ('assistant', 'assistant', ('text',))
Shell cwd was reset to D:\College\sem 5\SE_mini_project
```

Action: Write - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\chat_pdf.py

```
"""Render the exported Claude Code session as a PDF: prompts, replies, actions."""
import json
import re
import sys
from xml.sax.saxutils import escape

from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
from reportlab.lib.units import cm
from reportlab.platypus import Paragraph, Preformatted, SimpleDocTemplate, Spacer

src, out = sys.argv[1:3]
base = getSampleStyleSheet()
title = base["Title"]
body = ParagraphStyle("b", parent=base["BodyText"], fontSize=10, leading=14)
who = ParagraphStyle("w", parent=base["Heading4"], spaceBefore=10, spaceAfter=2)
act = ParagraphStyle("a", parent=body, fontSize=8.5, leading=11, textColor="#555555")
code = ParagraphStyle("c", fontName="Courier", fontSize=7.5, leading=9.5, leftIndent=8, spaceAfter=4)

EMAIL = re.compile(r"[\w.+-]+@[\w-]+\.[\w.-]+")


def clean(text):
    text = re.sub(r"<system-reminder>.*?</system-reminder>", "", text, flags=re.S)
    text = EMAIL.sub("[email removed]", text)
    return text.encode("ascii", "replace").decode().replace("?**", "**").strip()


def para(text):
    text = escape(clean(text))
```

```
    text = re.sub(r"\*\*(.+?)\*\*", r"<b>\1</b>", text)
    text = re.sub(r"`([^`]+)`", r'<font face="Courier">\1</font>', text)
    return [Paragraph(p.replace("\n", "<br/>"), body) for p in text.split("\n\n") if p.strip()]


def wrap(text, width=105):
    lines = []
    for line in clean(text).splitlines():
        while len(line) > width:
            lines.append(line[:width])
            line = "    " + line[width:]
        lines.append(line)
    return "\n".join(lines)


story = [
    Paragraph("Lab 4: VibeCoding Chat History", title),
    Paragraph("Tool: Claude Code (Claude Opus 5.5) &bull; Student: Pranav, PES1UG24CS466 &bull; Repo worked on:
SETAPESU26/35_qbert", body),
    Paragraph("Exported from the Claude desktop app session. Shows every prompt, reply and action, plus the exact "
              "code edits. Raw tool output and internal reasoning are omitted, and email addresses are removed.",
act),
    Spacer(1, 0.3 * cm),
]

for line in open(src, encoding="utf-8"):
    d = json.loads(line)
    if d.get("type") not in ("user", "assistant") or d.get("isMeta"):
        continue
    content = d["message"]["content"]
    if d["type"] == "user":
        texts = [content] if isinstance(content, str) else [b["text"] for b in content if b.get("type") == "text"]
        for t in texts:
            if clean(t):
                story += [Paragraph("Student prompt", who)] + para(t)
        continue
    for b in content:
        if b.get("type") == "text" and clean(b["text"]):
            story += [Paragraph("Claude", who)] + para(b["text"])
        elif b.get("type") == "tool_use":
            name, inp = b["name"], b["input"]
            if name.endswith("update_plan") or name == "ToolSearch":
                continue
            label = inp.get("description") or inp.get("file_path") or inp.get("session_id") or ""
            story.append(Paragraph(escape(clean(f"Action: {name.split('__')[-1]} - {label}")), act))
            if name == "Edit":
                story.append(Preformatted(wrap("- " + inp["old_string"].replace("\n", "\n- ")), code))
                story.append(Preformatted(wrap("+ " + inp["new_string"].replace("\n", "\n+ ")), code))

SimpleDocTemplate(out, pagesize=A4, leftMargin=2 * cm, rightMargin=2 * cm, topMargin=2 * cm, bottomMargin=2 * cm,
                  title="Lab 4 VibeCoding Chat History").build(story)
```

Result:
File created successfully at: C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\chat_pdf.py (file state is current in your context — no need to Read it back)

Action: Write - D:\College\sem 5\SE_mini_project\Lab-4\README.md

```
# Lab 4: VibeCoding (Q*bert Repair)

**Student:** Pranav &bull; **SRN:** PES1UG24CS466
**Assigned repo:** [SETAPESU26/35_qbert](https://github.com/SETAPESU26/35_qbert)
**AI tool used:** Claude Code

## Deliverables

| Deliverable | File |
| --- | --- |
| Video before changes | [before_video.mp4](before_video.mp4) |
| Video after changes | [after_video.mp4](after_video.mp4) |
| Updated code | [35_qbert/game.py](35_qbert/game.py) |
| Chat history | [Lab_4_Chat_History.pdf](Lab_4_Chat_History.pdf) |

## What changed in `game.py`

| Task | Change |
| --- | --- |
| 1. Projection bug | `cube_center` used `row // 2`, which rounds odd rows and skews the pyramid. It now uses `row /
2`. |
| 2. `cube_palette(level)` | Returns one of four 3-colour palettes, repeating after level 4. Level 1 keeps the
default colours. |
| 3. `on_cube_completed(cell)` | Starts a 0.35s white flash on the cube that was just completed. |
| 4. `bonus_life_threshold()` | Returns `1000`, so a life is awarded at every 1000 points. |

The flash also needed small changes outside the hook: `flashes.clear()` in `Game.reset`, a countdown at the top of
`Game.update`, and a colour blend in `Game.draw_cube`. The countdown runs before the state check so the last cube's
flash still fades on the level-clear screen.
```

```
## What the videos show

Both videos are 10 seconds long. They were captured by a script that ran the real `game.py` and saved its frames,
with a simple bot pressing the keys instead of a person.

- **Before:** the original code, with rows drifting sideways so the pyramid is skewed.
- **After:** the symmetric pyramid and cube flashes, a bonus life at 1000 points (1.2s), the level clear (7.7s), the
level 2 palette (8.5s), and a second bonus life at 2000 points (9.6s). This clip is a continuous 10-second part of a
longer run.

## Run it

```bash
pip install pygame
python 35_qbert/game.py
```

Controls: arrow keys to hop diagonally, `R` to reset, Space to continue after clearing a level.
```

Result:
```
File created successfully at: D:\College\sem 5\SE_mini_project\Lab-4\README.md (file state is current in your context
— no need to Read it back)
```

Action: Edit - D:\College\sem 5\SE_mini_project\README.md

```
- - [Full Deliverables Directory Listing](#-repository-deliverables-index)
+ - [Lab 4: VibeCoding (Q*bert Repair)](#-lab-4-vibecoding-qbert-repair)
+ - [Full Deliverables Directory Listing](#-repository-deliverables-index)
```

Result:
```
The file D:\College\sem 5\SE_mini_project\README.md has been updated successfully.
```

Action: Edit - D:\College\sem 5\SE_mini_project\README.md

```
- ---
-
- ##    Repository Deliverables Index
-
- ```
- .
- ├── README.md                                    # Master Laboratory Portfolio (Labs 1, 2, and 3)
-

+ ---
+
+ ##   Lab 4: VibeCoding (Q*bert Repair)
+
+ Fixed the pyramid projection bug and added the three features from the assigned repo, [SETAPESU26/35_qbert](https://
github.com/SETAPESU26/35_qbert), using Claude Code. The before and after videos, updated code, and chat history are
in [Lab-4](Lab-4/README.md).
+
+ ---
+
+ ##    Repository Deliverables Index
+
+ ```
+ .
+ ├── README.md                                    # Master Laboratory Portfolio (Labs 1 to 4)
+ ├── Lab-4/                                       # Lab 4 videos, updated code, and chat history
+
```

Result:
```
The file D:\College\sem 5\SE_mini_project\README.md has been updated successfully.
```

Action: Bash - Build chat PDF, assemble Lab-4, and verify contents

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; R="/d/
College/sem 5/SE_mini_project"; PY="$S/venv/Scripts/python.exe"; cd "$S" && "$PY" -I chat_pdf.py "$S/export/fd18d233-
152a-403c-8ec0-4d28662df9a4.jsonl" "$R/Lab-4/Lab_4_Chat_History.pdf" && mkdir -p "$R/Lab-4/35_qbert" && cp "$R/Lab4/
35_qbert/game.py" "$R/Lab4/35_qbert/README.md" "$R/Lab4/35_qbert/.gitignore" "$R/Lab-4/35_qbert/" && cp out/before.
mp4 "$R/Lab-4/before_video.mp4" && cp out/after.mp4 "$R/Lab-4/after_video.mp4"; cd "$R"; echo '=== game diff vs
upstream ==='; git -C Lab4/35_qbert diff --stat; echo '=== final check on Lab-4 copy ==='; "$PY" -I "$S/check_game.
py" "Lab-4/35_qbert/game.py" 2>&1 | tail -1; echo '=== pdf ==='; PYTHONIOENCODING=utf-8 python -c "
import pypdf,re
r=pypdf.PdfReader('Lab-4/Lab_4_Chat_History.pdf')
t=' '.join(p.extract_text() for p in r.pages)
print('pages',len(r.pages),'chars',len(t))
print('emails',re.findall(r'[\w.+-]+@[\w-]+\.\w+',t)[:5])
print('tokens',re.findall(r'gh[opsu]_[A-Za-z0-9]{6,}',t)[:5])
print(t[:700])
"; echo '=== status ==='; git status -s; du -sh Lab-4
```

Result:
```
=== game diff vs upstream ===
 game.py | 26 ++++++++++++++++++++-----
 1 file changed, 21 insertions(+), 5 deletions(-)
=== final check on Lab-4 copy ===
ALL CHECKS PASSED
=== pdf ===
pages 2 chars 5244
emails []
```

```
tokens []
Lab 4: VibeCoding Chat History
Tool: Claude Code (Claude Opus 5.5)   Student: Pranav, PES1UG24CS466   Repo worked on:
SETAPESU26/35_qbert
Exported from the Claude desktop app session. Shows every prompt, reply and action, plus the exact code edits. Raw
tool
output and internal reasoning are omitted, and email addresses are removed.
Student prompt
@"D:\College\sem 5\SE_mini_project\Lab_4_VibeCoding_Student_handout.pdf"
check it this is already implemented or do i need to do this and the reference lab to which u need to
commit to is https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466
and for the lab word use https://github.com/SETAPESU26/35_qbert tell me if is finished if it is fi
=== status ===
 M README.md
?? Lab-4/
?? Lab4/
?? Lab_4_VibeCoding_Student_handout.pdf
713K    Lab-4
```

Claude

The PDF is only 2 pages, which looks short for what it claims to cover. I'll check it really includes the replies and code edits before anything gets committed.

Action: Bash - Print the chat PDF body and compare with source

```
cd "/d/College/sem 5/SE_mini_project"; PYTHONIOENCODING=utf-8 python -c "
import pypdf
r=pypdf.PdfReader('Lab-4/Lab_4_Chat_History.pdf')
t='\n'.join(p.extract_text() for p in r.pages)
print(t[700:])
"; S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad";
echo '=== source counts ==='; "$S/venv/Scripts/python.exe" -I - "$S/export/fd18d233-152a-403c-8ec0-4d28662df9a4.
jsonl" <<'EOF'
import json,sys,collections
c=collections.Counter(); seen=set()
for l in open(sys.argv[1],encoding='utf-8'):
    d=json.loads(l)
    if d.get('type')!='assistant': continue
    for b in d['message']['content']:
        c[b.get('type')+':'+b.get('name','')]+=1
print(c)
EOF
```

```
Result:
nished
push it to the repo otherwise finish it and then commit it
Claude
I'll start by reading the handout and checking what's already in Lab4/, plus the state of both repos.
Action: Read - D:\College\sem 5\SE_mini_project\Lab_4_VibeCoding_Student_handout.pdf
Action: Bash - Show remotes, branches, and Lab4 file listing
Action: Bash - Check gh auth and inspect both GitHub repos
Action: Bash - Extract handout text with a Python PDF library
Action: Bash - Inspect local qbert clone history and README
Action: Bash - Check branches, PRs, forks, and permissions on qbert repo
Action: Bash - Extract handout text using UTF-8 output
Action: Read - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py
Action: Bash - See how the existing PR submits and check pygame
Action: Bash - Review repo layout conventions and README structure
Action: Bash - Create scratch venv with game and video packages
Action: Edit - D:\College\sem 5\SE_mini_project\Lab4\35_qbert\game.py
 - KEY_HOPS = {pygame.K_LEFT: (-1, -1), pygame.K_UP: (-1, 0), pygame.K_DOWN: (1, 0), pygame.K_RIGHT: (1, 1
    )}
-
-
- def cube_palette(level):
-     """Return a list of TARGET + 1 (r, g, b) colours for the cube stages, or None for the default."""
-     pass
-
-
- def on_cube_completed(cell):
-     """Called when a cube first reaches its target colour; add a flash, sound, or bonus here."""
-     pass
-
-
- def bonus_life_threshold():
-     """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
-     pass
-
-
- def cube_center(row, col):
-     return pygame.Vector2(WIDTH / 2 + (col - row // 2) * CUBE_W, 90 + row * CUBE_H)
+ KEY_HOPS = {pygame.K_LEFT: (-1, -1), pygame.K_UP: (-1, 0), pygame.K_DOWN: (1, 0), pygame.K_RIGHT: (1, 1
    )}
+ PALETTES = [
+     DEFAULT_PALETTE,
+     [(150, 90, 200), (235, 120, 170), (255, 225, 110)],
```

```
+     [(200, 80, 70), (240, 160, 60), (130, 230, 220)],
+     [(70, 130, 110), (170, 200, 80), (245, 245, 200)],
+ ]

+ FLASH_TIME = 0.35
+ flashes = {}
+
+
+ def cube_palette(level):
+     """Return a list of TARGET + 1 (r, g, b) colours for the cube stages, or None for the default."""
+     return PALETTES[(level - 1) % len(PALETTES)]
+
+
+ def on_cube_completed(cell):
+     """Called when a cube first reaches its target colour; add a flash, sound, or bonus here."""
+     flashes[cell] = FLASH_TIME
+
[... 50 more lines of output not shown ...]
```

Action: Bash - Search both transcript files for assistant replies

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; cd "$S/
export"; ls -la; "$S/venv/Scripts/python.exe" -I - <<'EOF'
import json,collections,glob
for f in ['transcript.jsonl','fd18d233-152a-403c-8ec0-4d28662df9a4.jsonl']:
    c=collections.Counter(); texts=[]
    for l in open(f,encoding='utf-8'):
        d=json.loads(l)
        c[d.get('type')]+=1
        m=d.get('message') or d
        ct=m.get('content') if isinstance(m,dict) else None
        if isinstance(ct,list):
            for b in ct:
                if isinstance(b,dict) and b.get('type')=='text' and (m.get('role')=='assistant' or d.get('type') in(
'assistant','api-request-blob')):
                    texts.append((d.get('type'),m.get('role'),b['text'][:70].replace('\n',' ')))
    print(f,dict(c.most_common(8)))
    seen=set()
    for t in texts:
        if t[2] not in seen and t[1]=='assistant':
            seen.add(t[2]); print('  ',t)
EOF
```

```
Result:
total 2712
drwxr-xr-x 1 sripr 197609       0 Nov  5 20:09 .
drwxr-xr-x 1 sripr 197609       0 Nov  5 20:09 ..
drwxr-xr-x 1 sripr 197609       0 Nov  5 20:09 fd18d233-152a-403c-8ec0-4d28662df9a4
-rw-r--r-- 1 sripr 197609 1196081 Nov  5 20:09 fd18d233-152a-403c-8ec0-4d28662df9a4.jsonl
-rw-r--r-- 1 sripr 197609     932 Nov  5 20:09 local-session-state.json
-rw-r--r-- 1 sripr 197609  361956 Nov  5 20:09 metadata.json
-rw-r--r-- 1 sripr 197609 1196081 Nov  5 20:09 transcript.jsonl
transcript.jsonl {'assistant': 35, 'user': 26, 'attachment': 20, 'last-prompt': 12, 'atis-latch': 12, 'custom-title':
11, 'api-request-blob': 10, 'agent-name': 10}
   ('assistant', 'assistant', "I'll start by reading the handout and checking what's already in `Lab4")
fd18d233-152a-403c-8ec0-4d28662df9a4.jsonl {'assistant': 35, 'user': 26, 'attachment': 20, 'last-prompt': 12, 'atis-
latch': 12, 'custom-title': 11, 'api-request-blob': 10, 'agent-name': 10}
   ('assistant', 'assistant', "I'll start by reading the handout and checking what's already in `Lab4")
Shell cwd was reset to D:\College\sem 5\SE_mini_project
```

Action: Edit - C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\chat_pdf.py

```
-     Paragraph("Exported from the Claude desktop app session. Shows every prompt, reply and action, plus the exact "
-               "code edits. Raw tool output and internal reasoning are omitted, and email addresses are removed.",
act),
+     Paragraph("Generated from the Claude desktop app's session export, taken after the code and videos were
finished "
+               "and before the final commit. Shows the prompt, the actions taken and the exact code edits. The
export "
+               "kept only the first of Claude's written replies, so later replies do not appear. Raw tool output and
"
+               "internal reasoning are omitted, and email addresses are removed.", act),
```

```
Result:
The file C:\Users\sripr\AppData\Local\Temp\claude\b612d014132b835b\fd18d233-152a-403c-8ec0-4d28662df9a4\scratchpad\
chat_pdf.py has been updated successfully. (file state is current in your context — no need to Read it back)
```

Action: Bash - Regenerate PDF, commit deliverables, push, verify on GitHub

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; R="/d/
College/sem 5/SE_mini_project"; cd "$S" && "$S/venv/Scripts/python.exe" -I chat_pdf.py "$S/export/fd18d233-152a-403c-
8ec0-4d28662df9a4.jsonl" "$R/Lab-4/Lab_4_Chat_History.pdf" && cd "$R" && git add Lab-4 README.md
Lab_4_VibeCoding_Student_handout.pdf && echo '=== staged ===' && git diff --cached --stat && git ls-files -s Lab-4 |
awk '{print $1, $4, $5}' && git commit -q -F - <<'EOF'
Add Lab 4 deliverables: Q*bert fix, features, videos, and chat history

Fix the pyramid projection bug and implement cube_palette,
on_cube_completed, and bonus_life_threshold in the assigned 35_qbert game.
```

```
Add before and after videos, the chat history PDF, and a Lab 4 README.

Co-Authored-By: Claude Opus 5.5 <[email removed]>
EOF
git push origin main 2>&1 | tail -3; echo '=== verify ==='; git status -sb | head -5; git log --oneline -2; gh api
'repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/contents/Lab-4' --jq '.[] | .type + " " + .name + " " + (.
size|tostring)'; gh api 'repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/contents/Lab-4/35_qbert' --jq '.[] | .
type + " " + .name + " " + (.size|tostring)'; gh api repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/contents/Lab-
4/35_qbert/game.py --jq .content | base64 -d | grep -n 'row / 2\|return 1000'
```

```
Result:
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Lab-4/Lab_4_Chat_History.pdf', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'Lab-4/README.md', LF will be replaced by CRLF the next time Git touches it
=== staged ===
 Lab-4/35_qbert/.gitignore            |   1 +
 Lab-4/35_qbert/README.md             |  91 ++++++++++++
 Lab-4/35_qbert/game.py               | 244 ++++++++++++++++++++++++++++++++++
 Lab-4/Lab_4_Chat_History.pdf         | 105 ++++++++++++++
 Lab-4/README.md                      |  41 ++++++
 Lab-4/after_video.mp4                | Bin 0 -> 351433 bytes
 Lab-4/before_video.mp4               | Bin 0 -> 327044 bytes
 Lab_4_VibeCoding_Student_handout.pdf | Bin 0 -> 71355 bytes
 README.md                            |  10 +-
 9 files changed, 491 insertions(+), 1 deletion(-)
100644 Lab-4/35_qbert/.gitignore
100644 Lab-4/35_qbert/README.md
100644 Lab-4/35_qbert/game.py
100644 Lab-4/Lab_4_Chat_History.pdf
100644 Lab-4/README.md
100644 Lab-4/after_video.mp4
100644 Lab-4/before_video.mp4
To https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466.git
   aca8600..e90e104  main -> main
=== verify ===
## main...origin/main
?? Lab4/
e90e104 Add Lab 4 deliverables: Q*bert fix, features, videos, and chat history
aca8600 Add Lab 2 and Lab 3 deliverables: Agile backlog & sprint simulation, UML component diagram, and architectural
justification
dir 35_qbert 0
file Lab_4_Chat_History.pdf 5779
file README.md 1990
file after_video.mp4 351433
file before_video.mp4 327044
file .gitignore 13
file README.md 3981
file game.py 8851
33:    return 1000
37:    return pygame.Vector2(WIDTH / 2 + (col - row / 2) * CUBE_W, 90 + row * CUBE_H)
```

Action: Bash - Download pushed PDF and verify it opens intact

```
S="/c/Users/sripr/AppData/Local/Temp/claude/b612d014132b835b/fd18d233-152a-403c-8ec0-4d28662df9a4/scratchpad"; R="/d/
College/sem 5/SE_mini_project"; cd "$S"; gh api repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/contents/Lab-4/
Lab_4_Chat_History.pdf --jq .content | base64 -d > remote_chat.pdf; ls -l remote_chat.pdf "$R/Lab-4/
Lab_4_Chat_History.pdf"; cmp remote_chat.pdf "$R/Lab-4/Lab_4_Chat_History.pdf" && echo IDENTICAL;
PYTHONIOENCODING=utf-8 python -c "
import pypdf
r=pypdf.PdfReader('remote_chat.pdf',strict=True)
print('pages',len(r.pages)); print(r.pages[0].extract_text()[:420])
"; for v in before after; do gh api repos/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/git/trees/main?recursive=1 --
jq ".tree[] | select(.path==\"Lab-4/${v}_video.mp4\") | .path + \" \" + (.size|tostring)"; done
```

```
Result:
-rw-r--r-- 1 sripr 197609 5779 Nov  5 20:11 /d/College/sem 5/SE_mini_project/Lab-4/Lab_4_Chat_History.pdf
-rw-r--r-- 1 sripr 197609 5779 Nov  5 20:11 remote_chat.pdf
IDENTICAL
pages 2
Lab 4: VibeCoding Chat History
Tool: Claude Code (Claude Opus 5.5)   Student: Pranav, PES1UG24CS466   Repo worked on:
SETAPESU26/35_qbert
Generated from the Claude desktop app's session export, taken after the code and videos were finished and before the
final
commit. Shows the prompt, the actions taken and the exact code edits. The export kept only the first of Claude's
written
replies, so later replies do not appea
Lab-4/before_video.mp4 327044
Lab-4/after_video.mp4 351433
Shell cwd was reset to D:\College\sem 5\SE_mini_project
```

Action: Wrap-up summary

```
Lab 4 is finished and pushed to main on your personal repo as commit e90e104, in the Lab-4 folder. Headless game
checks passed, but nobody has played it by hand. The videos are bot-driven and the chat PDF has only one Claude reply,
 so check both before submitting.
```

```
Result:
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer
```

```
to the user: write it now, as the last thing you say this turn.
```

Claude

Lab 4 wasn't finished, so I completed it and pushed it to your lab repo (https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/tree/main/Lab-4) (commit `e90e104`, on `main`). `game.py` was still the untouched starter code, and the videos and chat history didn't exist yet.

I didn't push anything to `SETAPESU26/35_qbert` or open a PR there, because the handout forbids PRs and your account only has read access.

What changed in `game.py`

| Task | Change |

|---|---|

| 1. Pyramid bug | `row // 2` is now `row / 2`, so the pyramid is symmetric |

| 2. `cube_palette` | 4 colour palettes that change each level and repeat after level 4 |

| 3. `on_cube_completed` | The finished cube flashes white for 0.35s |

| 4. `bonus_life_threshold` | Returns `1000`, so you get an extra life every 1000 points |

My automated checks passed for all four tasks and the README's expected behaviour, including falling, Coily chasing, winning and game over. They ran without a window, so nobody has played the game by hand yet.

Check before you submit

1. **The videos weren't recorded by a person.** A script ran the real game with a bot pressing the keys and saved the frames. The handout says to record gameplay, so you may want to record your own; `Lab-4/README.md` states how these were made. The after video is a 10-second part of a longer run, chosen to show every feature.

2. **The chat history is incomplete.** The app's export kept only my first written reply, so `Lab_4_Chat_History.pdf` has your prompt, that reply, the actions taken and the exact code edits. The PDF says this at the top. It also stops before the final commit and leaves out raw tool output, which contained other people's emails. The full export is at `C:\Users\sripr\Downloads\session-export-1793889569964.zip`.

3. **This may not match what the lab asks for.** The handout wants you to write prompts yourself and add the features one by one, but this chat is a single request that I did all at once. It's also an individual assignment, so check that this is acceptable. If not, redo the prompts yourself and replace the PDF.

Your local folders

The pushed work is in `Lab-4/`, the folder name the handout asks for. Your original `Lab4/35_qbert` clone is still on your computer, untracked and not pushed. It has the same fixed `game.py`, so you can delete it if you don't need it.

I also added a Lab 4 section to your main `README.md` and committed the handout PDF, as you did for Labs 2 and 3. To play the game yourself, run `pip install pygame` first, since it wasn't installed in your main Python.

Student

```
commit it to the git hub repo
```

Claude

It should already be up there from the last step, so I'll confirm GitHub matches your local copy and that nothing is left uncommitted.

Action: Bash - Compare local and GitHub commits, check uncommitted work

```
cd "/d/College/sem 5/SE_mini_project" && git fetch origin 2>&1 | tail -1; git status -sb; echo "local:  $(git rev-parse HEAD)"; echo "github: $(git ls-remote origin refs/heads/main | cut -f1)"; git log --oneline -1; echo '--- Lab4 clone vs Lab-4 ---'; cmp Lab4/35_qbert/game.py Lab-4/35_qbert/game.py && echo 'game.py identical'
```

```
Result:
## main...origin/main
?? Lab4/
local:  e90e10435b1013c63bac546b608096f6eb102ba4
github: e90e10435b1013c63bac546b608096f6eb102ba4
e90e104 Add Lab 4 deliverables: Q*bert fix, features, videos, and chat history
--- Lab4 clone vs Lab-4 ---
game.py identical
```

Claude

Nothing new needed committing: Lab 4 is already on GitHub. Your local `main` and GitHub are on the same commit, `e90e104`, so everything is in the `Lab-4` folder (https://github.com/Pranav-146/LAB1_Activity_Pranav_PES1UG24CS466/tree/main/Lab-4).

The only thing not committed is your local `Lab4/` folder, and that's on purpose. It's the original clone with its own `.git`, so committing it would store only a pointer rather than the code. Its `game.py` is identical to the pushed one, so nothing is missing.

If you meant `SETAPESU26/35_qbert`, I can't push there: your account has read-only access, and the handout says to keep your work in your own repo.